

You know from Chapter 12 that an electric current in a conductor can produce a magnetic field. Is the opposite also true? Can a magnetic field produce an electric current in a conductor? In 1831, Michael Faraday, an English scientist, proved that it could. This discovery led to many of the technologies that provide the electricity we use every day.

Discovery of Electromagnetic Induction

In Section 12.2 you learned that a constant electric current will produce a magnetic field, so it is logical to assume the opposite—that a constant magnetic field will produce an electric current in a conductor sitting in that constant magnetic field. It does not. Faraday discovered that in order to produce an electric current, the magnetic field needs to be continuously changing. He discovered **electromagnetic induction**, the production of electric current in a conductor within a changing magnetic field.

Induction means that one action causes another action to happen, often without direct contact. In his investigations, Faraday brought a permanent magnet near a conductor, but not in direct contact with it, and induced a current in the conductor. The electric current was produced only while the magnet was moving in the vicinity of the conductor. We call this an “induced current” because it is not an already existing current; it is formed by the action of the magnetic field moving along the conductor. These observations led Faraday to develop what is now known as the **law of electromagnetic induction**.

Law of Electromagnetic Induction

Any change in the magnetic field near a conductor induces a voltage in the conductor, causing an induced electric current in the conductor.

electromagnetic induction the production of electric current in a conductor moving through a magnetic field

law of electromagnetic induction a change in the magnetic field in the region of a conductor induces a voltage in the conductor, causing an induced electric current in the conductor

Mini Investigation

Faraday's Ring

Skills: Performing, Observing

SKILLS
HANDBOOK  A2.1

Electromagnetic induction can be investigated using a device containing two completely independent circuits. The primary circuit is connected to a source of electrical energy. The secondary circuit is connected only to a galvanometer.

Equipment and Materials: galvanometer; battery (or power supply); switch; alligator clip leads; 2 pieces of conducting wire; soft-iron ring

1. Construct a Faraday's ring apparatus as shown in **Figure 1**. Coil the conducting wire tightly around the ring. Be sure to coil the conducting wire the same number of times on each side of the ring.

2. Close the switch in the primary circuit and observe the galvanometer in the secondary circuit.
3. Open the switch in the primary circuit and observe the galvanometer in the secondary circuit.
 - A. What happened to the galvanometer when the primary circuit switch was closed? T/I
 - B. What happened to the galvanometer when the primary circuit switch was open? T/I
 - C. Was there a difference in the direction of the current in Steps 2 and 3? T/I

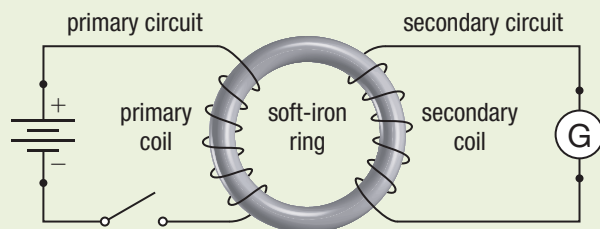


Figure 1 Faraday's ring

The implications of this discovery were extraordinary because for the first time electricity had been generated using only a magnet. Before Faraday's experiments, the only way to produce electrical energy was to use an electric cell or battery. Batteries are extremely useful, but they have disadvantages. Batteries operate for a limited amount of time, can be heavy and bulky, and produce only small electric potential differences. With Faraday's discovery, all that was needed was to find a way to move the magnet continuously, and the world would be able to produce electrical energy on a large scale without the limitations of batteries.

Electromagnetic Induction and Faraday's Ring

Faraday (**Figure 2**) investigated electromagnetic induction using a device that he built himself, the Faraday's ring, which you constructed in the Mini Investigation. Closing the switch in the primary circuit produces a constant electric current in the conducting wire. This constant electric current produces a magnetic field in the primary coil. The soft-iron ring enhances the strength of the magnetic field, and the ring itself becomes magnetized. This change of the magnetic field in the soft-iron ring (from zero to some value) induces a voltage and an electric current in the secondary circuit. However, once the magnetic field is stable and no longer changing, the electric current in the secondary circuit no longer exists. Remember that you need a changing magnetic field to induce an electric current.

When the switch is opened, the magnetic field in the primary coil disappears, because there is no longer an electric current. The magnetic field in the soft-iron ring collapses from maximum strength to zero. This change in the magnetic field causes an induced electric current in the opposite direction in the secondary circuit. Direct currents only produce electromagnetic induction for brief instants when the primary circuit is switched on or off.



Figure 2 Michael Faraday was born on September 22, 1791, in London, England. At 14, he apprenticed with a local bookbinder. He read many of the books that were being bound and developed an interest in electricity and chemistry. He was never formally trained as a scientist, yet he still published several papers in scientific journals.

Factors Affecting Electromagnetic Induction

Several factors determine the amount of electric current that can be produced by electromagnetic induction. Each of the following factors must be considered independently.

Coiled Conductor

In Section 12.4, you learned that by coiling a conductor you can create a magnetic field similar to that of a bar magnet. The magnetic fields from both sides of the loop interact to produce a more pronounced magnetic field in the centre of the loop. Similarly, with electromagnetic induction, a coiled conductor has more induced electric current in it than does a straight conductor.

The Number of Loops in the Coil

You know from Section 12.4 that increasing the number of loops in a coiled conductor, or solenoid, produces a stronger magnetic field for a given electric current. With electromagnetic induction, the number of loops in the coil is directly proportional to the magnitude of the electric current induced in the conductor for a given change in the magnetic field. So, the greater the number of loops in a coil, the more electric current can be induced for a given change in the magnetic field.

The Rate of Change of the Magnetic Field

There are two cases to consider here: a coiled conductor with a permanent magnet and a Faraday's ring apparatus. In the case of a coiled conductor with a permanent magnet, the more quickly you move the magnet into, or out of, the coil, the greater is the rate of change you cause in the magnetic field within the coil. A higher rate of change causes a larger induced electric current in the conductor.

In the second case of Faraday's ring, the more quickly you increase the current in the primary circuit, the greater is the rate of change you cause in the magnetic field in the coiled conductor and the soft-iron ring. The magnitude of the induced electric current in the secondary circuit is proportional to the rate of change of the magnetic field in the soft-iron ring.

The Strength of the Inducing Magnetic Field

The stronger the inducing magnetic field, the greater is the induced electric current. So, a stronger permanent magnet induces a greater electric current in a given coil. Similarly, in a Faraday's ring, a greater electric current in the primary circuit increases the strength of the magnetic field in the coiled conductor and soft-iron ring. This increases the induced electric current in the secondary circuit.

Applications of Electromagnetic Induction

To operate any electrical device, you rely on electrical power produced and supplied through generators and transformers—both of which rely on electromagnetic induction to operate. You will learn more about generators and transformers in Sections 13.4 and 13.5. Now we will look at three other applications of electromagnetic induction: induction cooking, metal detectors, and induction chargers.

Induction Cooking

Cooking food involves the transfer of thermal energy. In an electric stove, an electric current is directed into the element, which converts the electrical energy into thermal energy. That thermal energy is transferred by conduction into a metal pot. The pot needs to increase in temperature to then transfer thermal energy into food. The efficiency of this process is low because the stove element has to get hot, the pot has to get hot, and finally the food is heated. In the process, much thermal energy is lost to the environment.

Cooking using an induction cooker involves a rapidly changing magnetic field in the stove element (**Figure 3**), which induces an electric current in the pot. The electric current heats the pot because of the electrical resistance of the pot. Iron pots work better than copper or aluminum due to their higher electrical resistance. Insulating materials like glass will not work on an induction cooker. Cooking with an induction cooker is more efficient because it is a more direct transfer of thermal energy to the food. Another benefit is that the induction cooking surface does not get hot, so food that spills onto the cooking surface does not burn. Since the induction cooking surface is not hot, the food immediately stops heating up once the induction cooker is turned off.



Figure 3 An induction cooking surface

Metal Detectors

Electromagnetic induction is also used to detect metals. Metal detectors use a coil that generates a rapidly changing magnetic field. This magnetic field induces a current in any metal near it. The induced electric current in the detected metal also produces an induced magnetic field of its own. Sensitive measurements of the magnetic field near a metal detector are used to detect the induced magnetic field.

Metal detectors have many uses and have become quite common. They are used for humanitarian purposes to help locate buried bombs called land mines. Land mines buried during times of war are often not removed and innocent people are injured or killed when walking through areas where no land mine warnings exist. Metal detectors are also used for security purposes at airports. If you have been on a commercial flight, you have walked through one of these detectors. Security guards use handheld devices to detect any metal objects (**Figure 4**). In addition, metal detectors are used by hobbyists searching for potentially valuable metals that might be buried in the ground.



Figure 4 Metal detectors like this one use electromagnetic induction.

Induction Chargers

Electromagnetic induction can be used to charge low-power electronic devices such as electric toothbrushes, or even cellphones. The charger is plugged into a wall outlet. Both the charger and the device to be charged contain a wire coil. When the device is placed on the charger an electric current is induced in the coil inside the device. This induced current charges the internal battery of the device. The benefit of this type of charging is that you do not need wires to plug directly into the device you wish to charge. Also, in the case of an electric toothbrush, the lack of direct electrical contact allows you to safely charge the device even if it is wet. If you have several devices that need charging, you can use a single charging station and simply place the devices you wish to charge onto the station. The disadvantage of induction charging is that you require a special attachment to connect to the device you wish to charge. Also, charging by induction is less efficient because energy can be transformed into an unusable form during the transfer.

13.1 Summary

- The law of electromagnetic induction states that a change in the magnetic field in the region of a conductor induces a voltage in the conductor, causing an induced electric current in the conductor.
- Faraday's ring is a device that demonstrates electromagnetic induction. A current in the primary coil creates a magnetic field in the ring. The magnetic field in the ring then induces a current in the secondary coil.
- The amount of induced electric current can be increased by coiling the conductor, increasing the number of loops, increasing the rate of change of the magnetic field, or increasing the strength of the magnetic field.
- Electromagnetic induction is used in many technologies, including generators, transformers, induction cooking, metal detectors, and induction chargers.

13.1 Questions

1. A student demonstrates electromagnetic induction using a straight wire and a permanent magnet. The wire is part of a circuit that is connected to a galvanometer. What would you expect to happen in each of the following scenarios? **K/U**
 - (a) The magnet is placed on top of a stationary wire.
 - (b) The magnet is removed from the top of the stationary wire.
 - (c) The magnet is moved slowly over the top of the stationary wire.
 - (d) The magnet is moved quickly over the top of the stationary wire.
 - (e) The magnet is moved quickly back and forth over the stationary wire.
 - (f) The wire is moved quickly past the stationary magnet.
2. You need to demonstrate electromagnetic induction and wish to maximize the amount of induced current. Describe a design to accomplish this. **T/I C**
3. A glass cooking pot with an iron handle is placed on the cooking surface of an induction cooker. Describe the temperature of the glass and the handle after being on the induction cooker for some time. **K/U**
4. Could you design a non-metal detector that detects things other than metal and uses electromagnetic induction? Explain. **K/U**
5. Before going through a metal detector at an airport, you may have to remove your belt, empty your pockets, and remove your shoes. Explain why. **C A**
6. List some devices for which induction chargers could be used. Do you think the benefits outweigh the disadvantages? Explain your answer. **T/I A**